

JIN-WOO LEE

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EDUCATION

Gwangju Institute of Science and Technology

Aug. 2023 – Aug. 2025

M.S. in AI Convergence (Intelligent Robotics Program)

Advisor: Jiyeon Kang

Thesis: *Development of Adaptive Impedance Control Strategy for an Upper-Limb Cable-Driven Rehabilitation Robot*

Seoul National University of Science and Technology

Mar. 2017 – Aug. 2023

B.S. in Mechanical System Design Engineering

Advisor: Jung-Yup Kim

RESEARCH & PROFESSIONAL EXPERIENCE

AWEAR Lab, GIST

Gwangju, KR

Graduate Researcher | Lab Representative

Aug. 2023 – July. 2025

- Developed a cable-driven robotic platform for upper-limb rehabilitation with real-time force sensing and compliant control.
- Formulated and implemented real-time constrained optimization for cable tension allocation, maintaining feasible tension bounds while generating desired Cartesian forces.
- Developed an adaptive impedance controller with online adaptation of stiffness and damping profiles to modulate assistance and resistance based on user performance.
- Conducted human-subject experiments to validate the controller, demonstrating performance-adaptive assist-resist behavior while preserving range of motion and trajectory accuracy.

HRR Lab, SeoulTech

Seoul, KR

Undergraduate Researcher

Oct. 2021 – Dec. 2022

- Designed and built a lightweight 6-DoF manipulator using quasi-direct-drive (QDD) actuators for mobile manipulation.
- Implemented task-space torque control with gravity compensation to achieve compliant Cartesian motion.
- Developed a center-of-mass (CoM) position controller using a mass-weighted Jacobian derived from link masses and joint kinematics.
- Developed and integrated a ROS-based control system for both simulation validation and real-hardware implementation.

Samsung Electronics

Suwon, KR

Motor Control Engineer

July. 2025 – Present

- Contributed to the design and hardware debugging of a three-phase inverter for PMSM drives, including switching behavior, fault diagnosis, and dynamometer-based motor testing.
- Developed an automated lookup-table (LUT) generation tool for MTPA, field-weakening, and MTPV control of PMSM/IPMSM drives from motor parameters.

PUBLICATIONS

Jin-Woo Lee, Seonghun Jeong, and Jiyeon Kang, “Assist–Resist-as-Needed Robotic Control for Cable-Driven Bedside Training,” *2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.

Jin-Woo Lee, Sahil Kapahi, Souvik Poddar, and Jiyeon Kang, “Enhanced Feedforward Force Controller for Cyclic Disturbance Compensation in a Gait-Training Robot,” *IEEE Transactions on Medical Robotics and Bionics*, 2026, in press. doi: 10.1109/TMRB.2026.3722280.

Jin-Woo Lee and Jung-Yup Kim, “Design and Motion Control of Mobile Manipulator for Telepresence Using VR Headset,” *KSME 2022*, Jeju, Republic of Korea, Nov. 2022.

Yaeun Lee, **Jin-Woo Lee**, Nagyeong Ham, and Euseok Hwang, “Smart Fault Detection in Electric Vehicles Using Battery and Motor Operation Data Driven Deep Learning,” *2025 16th International Conference on Information and Communication Technology Convergence (ICTC)*, pp. 457–458, 2025.

ADDITIONAL ACTIVITIES

Senior KATUSA, Korean Augmentation to the U.S. Army
Served in a formal platoon leadership role and liaised between U.S. and Korean military leadership. Camp Humphreys | 2019

English Tutor, University Tutoring Program
SeoulTech | 2020

Encouragement Award, Invention Prototype Competition
Data-Driven Automotive Wheel Design SeoulTech | 2020

Encouragement Award, Super Local Project
Braille Education Device for the Visually Impaired SeoulTech | 2021

Co-Inventor, Korea Patent No. 10-2722143, granted
Gripper Detachment and Rotation Device SeoulTech | 2023

Selected Competitor, H-Mobility Robotics Hackathon
Selected from the robotics training cohort for a two-day pick-and-place robotics challenge. Hyundai Motor Company | 2023

Co-Inventor, Korea Patent Application No. 10-2025-0202284
Bedside Rehabilitation Exercise Device and Method GIST | 2025

Excellence Award, CXI New Employee Training
Selected as the outstanding trainee among 19 new employees. Samsung Electronics | 2025

Finalist & Presenter, Super Rookie Project
Selected for final-round presentation in an internal competition (top 10% of 300+ employees).
Robotic Arm Concept for Monitor Positioning Samsung Electronics | 2026

Robot Learning Project
Hands-on training in reinforcement learning for legged robots using Isaac Sim/Lab. Present

SKILLS

- **Robotics:** ROS, Gazebo, Isaac Sim/Lab
- **Programming:** Python, MATLAB, C/C++
- **Engineering Tools:** SolidWorks, LabVIEW, Git
- **Languages:** Korean (Native), English (Advanced, OPIc AL)